Parametrization of Peak Prediction Using Quantile Regression and Asymmetric Laplace Noise

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Abstract

In many signal processing applications, defining and finding peaks is an important part of the pipeline. Peak prediction can be a very challenging endeavor, especially when there is a lot of noise. In this work, the peak region is considered as a combination of a Gaussian trend with additive asymmetric Laplace noise, and the peak is modeled as the peak of the Gaussian trend.

We propose a criterion based on the weighted quantile which predicts the optimal number of observations necessary for the exact prediction of the peak's location. We analyze the effect of the signal-to-noise ratio on our prediction. We show that for small signal-to-noise ratio (a lot of noise), the prediction becomes inaccurate.

In this case, we propose a new idea called the entry time into the region of the peak and the exit time from the region of the peak. We demonstrate how to apply our new idea to real data and we analyze the effect of the choice of the objective function. As a numerical application, we consider the peak prediction of the daily infections during the first wave of COVID-19 in China, France, and Germany.

Keywords: Gaussian model, least absolute deviation, weighted median, entry time, region of the peak, daily infection.

Mathematics subject classification: 62M10, 60G35.

1 Introduction

Predicting peaks in time series data is crucial across various domains for proactive decision-making, risk management, resource allocation, and disaster preparedness. The ability to anticipate spikes in data can serve as an early warning system for events such as disease outbreaks in epidemiology, allowing public health officials to allocate resources and implement preventive measures effectively. In energy management, predicting peak electricity demand enables utility companies to adjust generation and distribution schedules, ensuring a reliable power supply

and preventing shortages. Financial analysts use peak prediction to manage investment risks by anticipating fluctuations in stock prices or market volatility. Additionally, environmental scientists rely on peak prediction to monitor pollution levels and natural disasters, facilitating disaster preparedness and response efforts. The examples justifying the usefulness of locating, measuring, and understanding these peaks are numerous.

Predicting peaks in time series data is a complex task due to several reasons. Firstly, the presence of noise and variability in the data makes it difficult to accurately identify and predict peaks. This noise can originate from various sources, such as measurement errors or random fluctuations in the data. Additionally, many real-world systems exhibit nonlinear dynamics, meaning that the relationship between variables is not strictly linear. Nonlinearities can lead to unexpected behaviors in the data, further complicating peak prediction. Moreover, time series data often contain complex patterns, such as multiple overlapping peaks or irregular intervals between peaks, making it challenging to develop accurate prediction models. In the literature, existing peak prediction methods often rely on specific assumptions about the data distribution or peak characteristics, which may not always hold true in practice, leading to inaccurate predictions (see for instance, Pei and Shaman (2017) and Tucker (2020)).

Popular forecasting models such as Autoregressive Integrated Moving Average (ARIMA) and Recurrent Neural Networks (RNN) often struggle to accurately define or forecast peak values in time series data. Despite their effectiveness in capturing general trends and patterns, these models may overlook or underestimate the occurrence of peaks due to their complexity and the inherent variability of peak events.

In the realm of peak prediction, there has been relatively little theoretical work compared to more generalized forecasting approaches. One notable contribution in this area is Tucker thesis Tucker (2020), which delved into the dynamics of trend, seasonality, and noise in time series data to predict peaks. Tucker research explored how the interplay between these components could be leveraged to identify and forecast peak events accurately. Inspired by previous studies Yu (2019), Tucker applied a combination of general prediction techniques and specialized peak prediction methods. He utilized the Peaks function available in Scipy's Signal module, which offers tools for detecting and analyzing peak values in data. Tucker thesis represents a significant step forward in the field of peak prediction, offering insights into the complex dynamics underlying peak events and proposing practical methods for their prediction. By building on existing theoretical frameworks and leveraging advanced computational tools, Tucker work contributes to the growing body of research aimed at improving the accuracy and reliability of peak forecasting in time series data.

We adopt in this work the same approach as in Dermoune et al. (2025) . We assume that the peak region is equal to $[T, h]$, where T represents the current date and h a specific time horizon. To model this hypothesis effectively, we consider our time series over the interval $[1, T]$ as a

composite of a Gaussian trend and noise. Our primary objective is to forecast the amplitude A, location L, and width S of the Gaussian trend based on observed data from the interval $[1, T]$. We introduce a criterion based on the weighted quantile technique, which enables us to determine the optimal number of observations required for accurately predicting the location L of the peak. Additionally, we conduct an analysis to assess the impact of the signal-to-noise ratio on our predictive accuracy. We show that in case of low signal-to-noise ratios, prediction accuracy diminishes. To address this challenge, we propose as in Dermoune et al. (2025) a novel concept termed the "peak region entry and exit times".

The aim of our study is to extend the work of Dermoune et al. (2025) to the quantile regression. First, we introduce our approach to selecting the minimization of the quantile regression and demonstrate, using simulated data, the number of observations required to accurately predict the location of the peak. This approach is based on the concept of the weighted quantile. Secondly, we establish that for high signal-to-noise ratios, the peak location prediction is correct. However, in cases of low `SNR` (indicating high noise levels), the accuracy of the prediction deteriorates. In these cases, we propose a new concept called "peak region entry time" and "peak region exit time". We illustrate this concept by parameterizing the entry time in the peak region and the exit time from the peak region by the tau-quantile when the data are the daily COVID-19 infections during the first wave in China Barmparis and Tsironis (2020), France and Germany.

The remainder of this paper is organized as follows. In Section 2 we present our proposed model. More precisely, we consider Gaussian trend with additive generalized Laplace noise. Section 8 is dedicated to application results to forecast the peak incidence of daily COVID-19 infections during the initial wave in China, France, and Germany.

2 Quantile-Based Estimation Methodology

We consider the problem of estimating the amplitude, location, and spread of a unimodal signal corrupted by noise. The model is:

$$I(t) = A \exp\left(-\frac{(t-L)^2}{S^2}\right) + e(t), \quad t = 1, \ldots, T,$$

where $A > 0$ is the amplitude, $L > 0$ the peak location, $S > 0$ the spread (scale), and $e(t)$ is additive noise. Given observations $(I(t))_{t=1}^{T}$, our goal is to estimate the underlying parameters using quantile regression, i.e., minimizing the asymmetric loss

$$\rho_\tau(u) = u \cdot (\tau - \mathbf{1}_{\{u<0\}}),$$

for a fixed quantile level $\tau \in (0, 1)$.

We define the quantile objective:

$$f_\tau(T, a, l, s) = \frac{1}{T} \sum_{t=1}^{T} \rho_\tau \left(I(t) - a \exp\left(-\frac{(t-l)^2}{s^2} \right) \right).$$

For fixed candidate values of (l, s), the optimal amplitude a minimizes $a \mapsto f_\tau(T, a, l, s)$. This minimizer is a weighted τ-quantile of a transformed version of the data, as described below.

2.1 Weighted Quantile Estimation of Amplitude

Let $w(t) = \exp\left(-\frac{(t-l)^2}{s^2}\right)$ be the weight associated with time point t, and define:

$$x(t) = I(t) \exp\left(\frac{(t-l)^2}{s^2} \right).$$

Then the optimal amplitude a is the weighted τ-quantile of $x(t)$ with weights $w(t)$, i.e.,

$$a = \arg\min_{a'} \sum_{t=1}^{T} w(t) \rho_\tau(a' - x(t)).$$

The proof of the following Proposition is the same as in the work of Novoselac (2020).

Proposition 1. *Let $(x(t), w(t))$ be a sequence of real values with positive weights $w(t) > 0$. Then, for any $\tau \in (0, 1)$, the weighted τ-quantile minimizes the weighted quantile loss:*

$$q_\tau = \arg\min_{a \in \mathbb{R}} \sum_{t=1}^{T} w(t)\, \rho_\tau(a - x(t)),$$

where $\rho_\tau(u) = u(\tau - \mathbf{1}\{u < 0\})$ is the quantile check function.

To compute q_τ, sort the values $x(t)$ in increasing order, obtaining a permutation π such that

$$x(\pi(1)) \leq x(\pi(2)) \leq \cdots \leq x(\pi(T)).$$

Define the reordered weights $w_\pi(t) = w(\pi(t))$. Let

$$k = \min \left\{ j \in \{1, \ldots, T\} : \sum_{t=1}^{j} w_\pi(t) \geq \tau \sum_{t=1}^{T} w(t) \right\}.$$

Then the weighted τ-quantile satisfies $q_\tau = x(\pi(k))$.

2.2 Quantile-Based Estimation Procedure

We define the following steps to jointly estimate the location $\hat{l}(T)$, amplitude $\hat{a}(T)$, and spread $\hat{s}(T)$:

Algorithm 1 Quantile-based estimation of peak parameters

Require: Time horizon T, upper bound $h > T$, quantile level $\tau \in (0, 1)$

1: For each candidate $l \in [1, h]$, find $s^*(l) = \arg\min_s f_\tau(T, a^*(l, s), l, s)$
2: Set $\hat{l}(T) = \arg\min_l f_\tau(T, a^*(l, s^*(l)), l, s^*(l))$
3: Set $\hat{s}(T) = s^*(\hat{l}(T))$
4: Set $\hat{a}(T) = a^*(\hat{l}(T), \hat{s}(T))$

Ensure: Estimated parameters $(\hat{l}(T), \hat{a}(T), \hat{s}(T))$

2.3 Remark on Asymmetry and Robustness

This method generalizes the classic Least Absolute Deviations (LAD) estimator, recovered at $\tau = 0.5$. By adjusting τ, we introduce asymmetry into the estimation loss, enabling the model to better adapt to skewed noise distributions. This makes the approach especially valuable in early-warning systems and epidemic peak detection where overestimation and underestimation have asymmetric consequences.

3 Quantile-Based Objective Function under Noise-Free Conditions

We first consider the idealized scenario where the signal is not corrupted by noise. The model of interest is given by

$$I(t) = A \exp\left(-\frac{(t-L)^2}{S^2}\right), \quad t = 1, \ldots, T,$$

where $A = 15000$, $L = 22$, $S = 10$, and $T \in \{2, 3, 10, 20, 21\}$. The quantile-based estimation procedure is implemented by minimizing the following objective function:

$$f_\tau(T, a, l, s) = \frac{1}{T} \sum_{t=1}^{T} \rho_\tau \left(I(t) - a \exp\left(-\frac{(t-l)^2}{s^2}\right)\right),$$

where $\rho_\tau(u) = u(\tau - \mathbf{1}_{\{u<0\}})$ denotes the asymmetric quantile loss. The amplitude a is computed via a weighted τ-quantile, as described in Section 2.

Since the goal is to predict the location L of the peak before it becomes observable, we restrict the minimization over candidate peak locations $l > T$. For each value of $\tau \in \{0.25, 0.5, 0.75, 0.95\}$, we compute and visualize the objective curve $l \mapsto f_\tau(T, a(T, l), l, s(T, l))$ for the selected values of T (see Figure 1).

Remark. For each fixed sample size T, we computed the value of the minimized objective function at the estimated parameters $(\hat{a}, \hat{l}, \hat{s})$. The results indicate that the lowest objective value is typically achieved for $\tau = 0.95$, suggesting that upper quantiles align particularly well with the noiseless signal structure. The only exception is at $T = 20$, where $\tau = 0.75$ yields a slightly smaller loss. This empirical observation may help guide quantile selection when prior

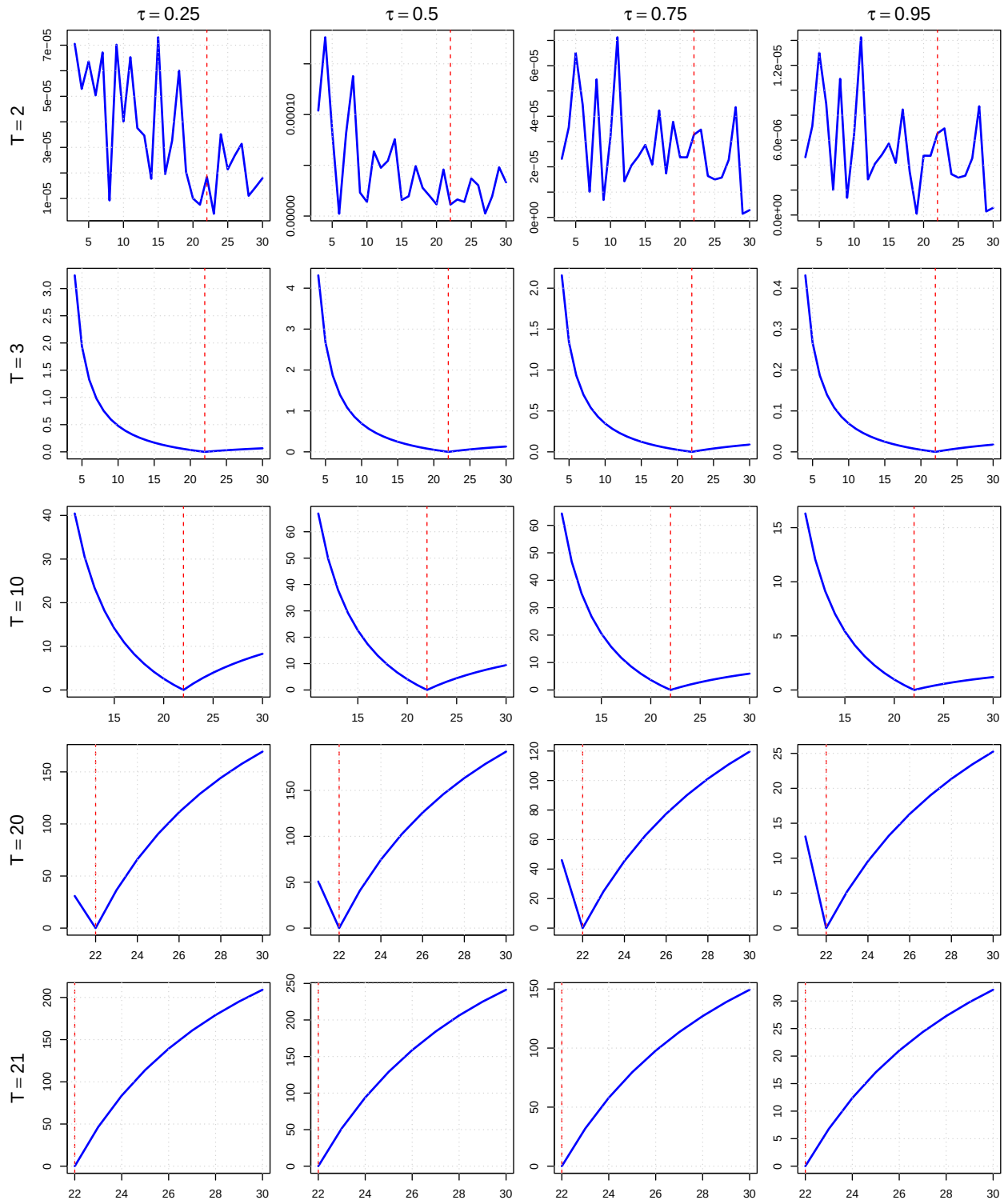


Figure 1: Objective function $l \mapsto f_\tau(T, a, l, s)$ computed for $\tau \in \{0.25, 0.5, 0.75, 0.95\}$ and $T \in \{2, 3, 10, 20, 21\}$, under a noise-free signal. The vertical dashed line indicates the true peak location $L = 22$, and the search is restricted to values $l > T$.

knowledge of signal shape or noise properties is limited.

4 Quantile-Based Estimation with τ-Aligned Noise

To analyze how quantile-aligned noise affects peak estimation, we introduce a unified noise model based on the *general asymmetric Laplace* (GAL) distribution. This flexible family allows both symmetric and asymmetric noise structures while maintaining the property that the τ-quantile of the noise is at zero.

4.1 Unified Noise Model: General Asymmetric Laplace

Let us start by announcing the following result.

Proposition 2. *Let $z \in (-1,1)$ and $b, \lambda > 0$. The probability distribution proportional to $x \in \mathbb{R} \to \exp(-\frac{(z-\lambda\frac{x}{b})^2}{2\lambda} - \frac{|x|}{\lambda b})$ converges in distribution to the probability distribution proportional to $x \in \mathbb{R} \to \exp(-\frac{|x|}{b}(1 - sgn(x)z)$ as $\lambda \to 0$. The latter distribution has the density*

$$\frac{(1-z)(1+z)}{2b}\exp\left(\frac{(1+z)x}{b}\right)\mathbf{1}_{[x<0]} + \frac{(1+z)(1-z)}{2b}\exp\left(-\frac{(1-z)x}{b}\right)\mathbf{1}_{[x>0]}.$$

The convergence in distribution means that $\int_a^b \exp(-\frac{(z-\lambda\frac{x}{b})^2}{2\lambda} - \frac{|x|}{\lambda b})\frac{dx}{c(z,b,\lambda)} \to \int_a^b \exp(-\frac{|x|}{b}(1 - sgn(x)z)\frac{dx}{c(z,b)}$ for each couple of real numbers $a < b$. Here $c(z,b,\lambda), c(z,b)$ are the constants of the normalizations (see Dermoune et al. (2017)).

Now we are going to solve the parameters of the model in Section 2 using τ-quantile regression with $\tau \in (0,1)$. To do that we need to assume that the noise e satisfies $\mathbb{P}(e < 0) = \tau$. We consider two symmetric cases of this model: 1) Shifted symmetric Laplace: obtained by shifting a standard Laplace so that its τ-quantile is at zero, and 2) the generalized Laplace noise e parametrized by the three parameters τ, z, b having the probability density function

$$f_{\tau,z,b}(x) = \begin{cases} \tau\dfrac{(1+z)}{b}\exp\left(\dfrac{(1+z)x}{b}\right), & x < 0, \\ (1-\tau)\dfrac{(1-z)}{b}\exp\left(-\dfrac{(1-z)x}{b}\right), & x \geq 0. \end{cases}$$

clearly this noise satisfies the condition $\mathbb{P}(e < 0) = \tau$ and its variance equals

$$\mathrm{Var}(e) = b^2\left[\frac{\tau}{(1+z)^2} + \frac{1-\tau}{(1-z)^2} - \frac{1}{4}\left(\frac{1-\tau}{1-z} - \frac{\tau}{1+z}\right)^2\right].$$

We obtain an asymmetric Laplace with $z = 0$: the GAL model reduces to a symmetric distribution centered at the τ-quantile. We analyze both constructions and later generalize to arbitrary values of z in Section 5.

4.2 Shifted Symmetric Laplace (Aligned at τ)

We now consider a heuristic construction in which the additive noise is drawn from a symmetric Laplace distribution and then shifted so that its τ-quantile is exactly zero. This ensures that the noise is statistically aligned with the asymmetric loss ρ_τ, i.e., $\mathbb{P}(e(t) < 0) = \tau$, even though the original noise is symmetric around zero.

More precisely, we define the noise as:

$$e(t) \sim \text{Laplace}(0, b) - q_\tau, \quad \text{with} \quad q_\tau = \begin{cases} b \log(2\tau) & \text{if } \tau < 0.5, \\ -b \log(2(1-\tau)) & \text{if } \tau \geq 0.5. \end{cases}$$

Here, q_τ is the theoretical τ-quantile of the Laplace distribution with scale b, and the subtraction guarantees that $\mathbb{P}(e(t) < 0) = \tau$.

Remark. *Although this construction achieves τ-alignment, it does not correspond to the general asymmetric Laplace (GAL) distribution introduced in Section 4.1. The shifted Laplace noise is obtained by translating a symmetric distribution, whereas GAL noise is inherently asymmetric in its density function and cannot be obtained by shifting a symmetric Laplace.*

We simulate observations as:

$$I(t) = A \exp\left(-\frac{(t-L)^2}{S^2}\right) + e(t),$$

where $A = 1$, $L = 22$, $S = 10$, and the scale parameter b is computed so that the signal-to-noise ratio equals $\text{SNR} = 1000$, using the variance of the aligned GAL noise:

$$b = \frac{A}{\text{SNR}} \cdot [\text{Var}_{\tau,z}(e)]^{-1/2}.$$

For each $\tau \in \{0.25, 0.5, 0.75, 0.95\}$ and $T \in \{4, 6, 8, 9, 10\}$, we evaluate the objective function:

$$f_\tau(T, l) := \min_s \frac{1}{T} \sum_{t=1}^{T} \rho_\tau \left(I(t) - a(T, l, s) \exp\left(-\frac{(t-l)^2}{s^2}\right)\right),$$

where $a(T, l, s)$ is computed as the weighted τ-quantile of the transformed observations $x(t) = I(t) \exp\left(\frac{(t-l)^2}{s^2}\right)$, with weights $w(t) = \exp\left(-\frac{(t-l)^2}{s^2}\right)$.

Figure 2 shows the resulting profiles $l \mapsto f_\tau(T, l)$ across different quantile levels and sample sizes. The red dashed line marks the true peak location $L = 22$.

4.3 Quantile-Based Estimation under Asymmetric Laplace Noise

In this subsection, we revisit the quantile-based estimation procedure using a noise model that is fully compatible with the asymmetric quantile loss function ρ_τ. Specifically, we assume that the additive noise follows the asymmetric Laplace distribution, which arises as the maximum

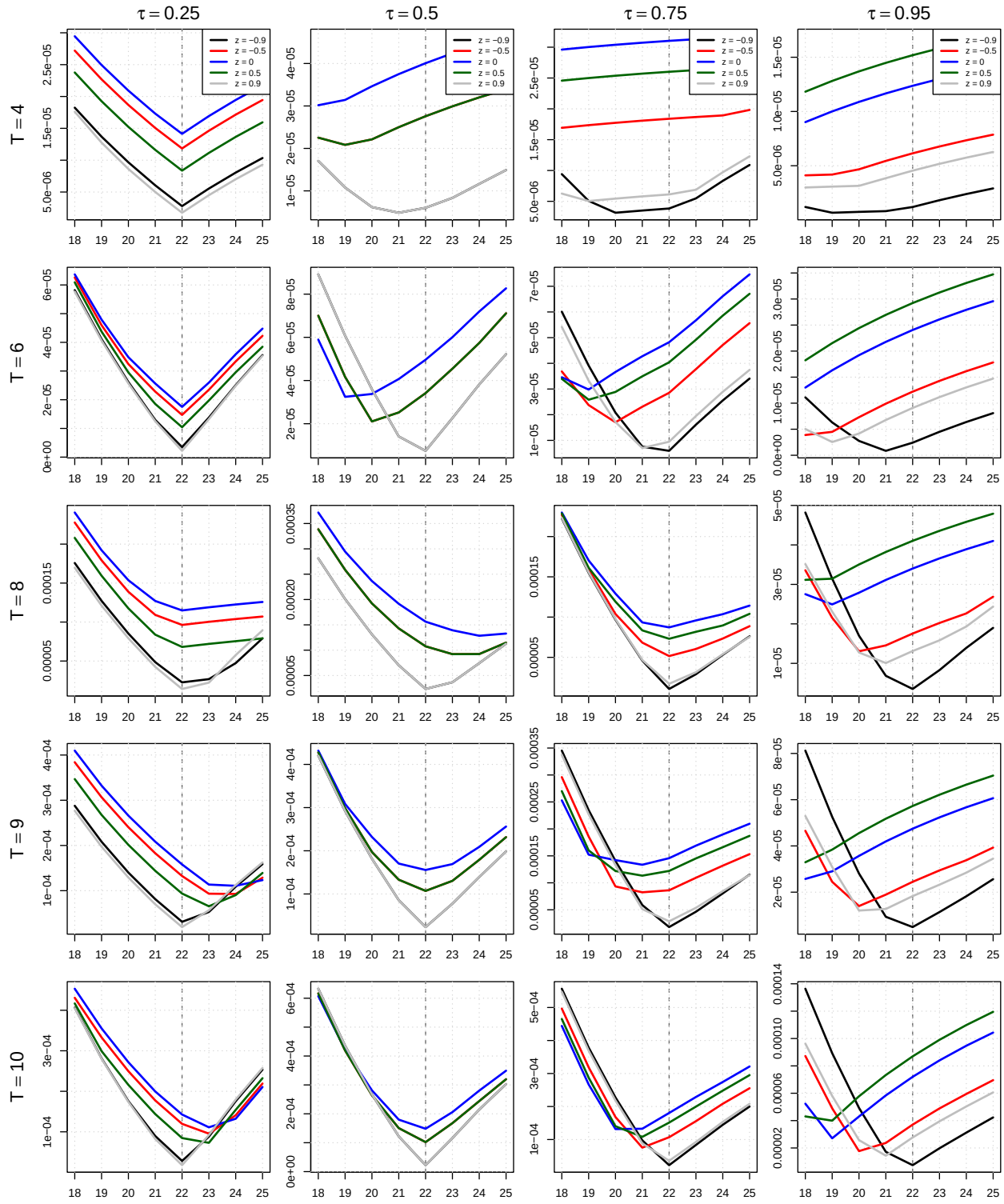


Figure 2: Objective profiles $l \mapsto f_\tau(T, a, l, s)$ under *shifted Laplace* noise, centered at the quantile level $\tau \in \{0.25, 0.5, 0.75, 0.95\}$, for varying time horizons $T \in \{4, 6, 8, 9, 10\}$. Each panel corresponds to a different pair (T, τ), and curves represent different skewness levels $z \in \{-0.9, -0.5, 0, 0.5, 0.9\}$. The dashed vertical line indicates the true peak location $L = 22$.

likelihood model for quantile regression. This distribution has density

$$f_{\tau,b}(x) = \begin{cases} \dfrac{\tau}{b} \exp\left(-\dfrac{\tau x}{b}\right), & x \geq 0, \\ \dfrac{1-\tau}{b} \exp\left(\dfrac{(1-\tau)x}{b}\right), & x < 0, \end{cases}$$

where $\tau \in (0,1)$ denotes the target quantile and $b > 0$ the scale parameter. This distribution ensures that $\mathbb{P}(e(t) < 0) = \tau$, in perfect alignment with the quantile loss minimization objective.

To ensure that $f_{\tau,b}(x)$ integrates to 1, one can verify that its normalizing constant is $\frac{\tau(1-\tau)}{b}$. Indeed, since $\rho_\tau(x) = x(\tau - \mathbf{1}\{x < 0\})$, the function $f_{\tau,b}(x) \propto \exp\left(-\frac{1}{b}\rho_\tau(x)\right)$ defines a valid probability density on $\mathbb{R}$, with

$$\int_{-\infty}^{+\infty} \exp\left(-\frac{1}{b}\rho_\tau(x)\right) dx = \int_{-\infty}^{0} \exp\left(\frac{(1-\tau)x}{b}\right) dx + \int_{0}^{+\infty} \exp\left(-\frac{\tau x}{b}\right) dx = \frac{b}{1-\tau} + \frac{b}{\tau}.$$

Hence the proper density is

$$f_{\tau,b}(x) = \frac{\tau(1-\tau)}{b} \exp\left(-\frac{1}{b}\rho_\tau(x)\right),$$

which is precisely the distribution used to simulate the noise in our experiments.

We replicate the simulation design used in the previous subsection: the true signal is $I(t) = A \exp\left(-\frac{(t-L)^2}{S^2}\right)$ with $A = 1$, $L = 22$, and $S = 10$, and the signal-to-noise ratio is set to $\text{SNR} = 1000$. For each $\tau \in \{0.25, 0.5, 0.75, 0.95\}$, we generate additive asymmetric Laplace noise with appropriate scale b ensuring that the variance of the noise matches the desired SNR level. The objective function $l \mapsto f_\tau(T, a, l, s)$ is computed for time horizons $T \in \{4, 6, 8, 9, 10\}$, and for each l in a discrete grid, we minimize over s and determine the optimal amplitude a as the weighted τ-quantile.

The resulting profiles of the objective function are displayed in Figure 3. The true peak location $L = 22$ is marked by a vertical dashed line. Each panel corresponds to a different pair (T, τ), and the curves illustrate the behavior of the loss function across different candidate peak locations. The red, blue, green, and grey lines correspond to varying levels of skewness z, introduced implicitly through the choice of τ.

Remark. A notable feature of Figure 3 is that all curves corresponding to different skewness levels $z \in \{-0.9, -0.5, 0, 0.5, 0.9\}$ fall exactly on top of each other. This occurs because the data were generated using the asymmetric Laplace density $f_\tau(x)$ that is exactly matched to the quantile loss function $\rho_\tau(x)$, i.e., the negative log-likelihood is proportional to the loss function being minimized.

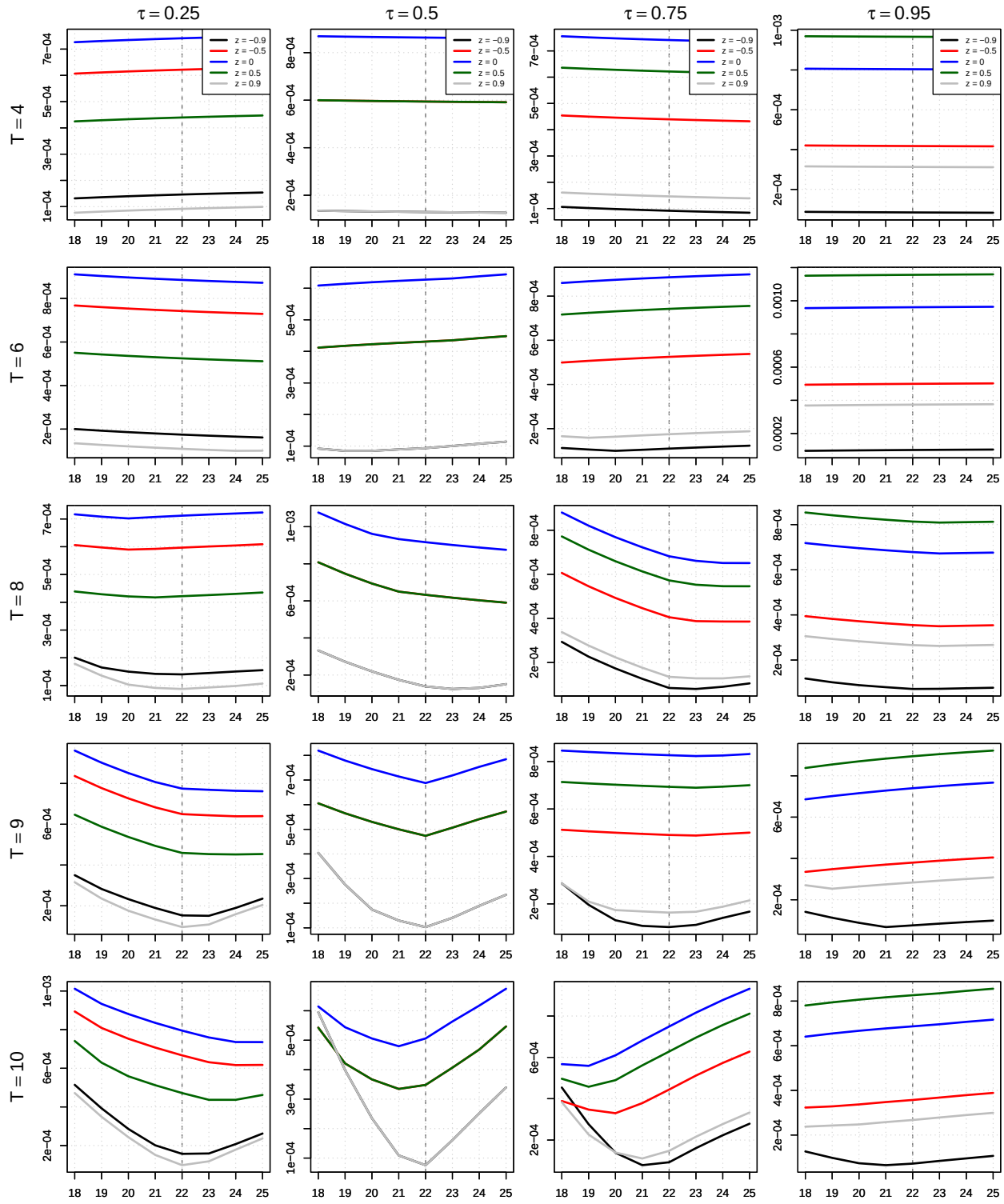


Figure 3: Objective function profiles $l \mapsto f_\tau(T, a, l, s)$ under *asymmetric Laplace* noise, for $\tau \in \{0.25, 0.5, 0.75, 0.95\}$ and $T \in \{4, 6, 8, 9, 10\}$. Each curve corresponds to a different skewness setting $z \in \{-0.9, -0.5, 0, 0.5, 0.9\}$, and the dashed vertical line indicates the true peak location $L = 22$.

4.4 Asymmetric Laplace Noise with $\tau = \frac{1-z}{2}$

We now consider the noise model derived in Proposition 2, in which the generalized asymmetric Laplace (GAL) kernel

$$x \mapsto \exp\left(-\frac{(z-\lambda x/b)^2}{2\lambda} - \frac{|x|}{\lambda b}\right)$$

converges (as $\lambda \to 0$) to a piecewise exponential distribution with skewness parameter $z \in (-1, 1)$. The limiting density is given by:

$$f_{z,b}(x) = \begin{cases} \dfrac{(1-z)(1+z)}{2b} \exp\left(\dfrac{(1+z)x}{b}\right), & x < 0, \\ \dfrac{(1-z)(1+z)}{2b} \exp\left(-\dfrac{(1-z)x}{b}\right), & x > 0, \end{cases}$$

which corresponds to an asymmetric Laplace distribution with skewness z and scale b, and has left tail probability $\mathbb{P}(e < 0) = \tau$, for $\tau = \frac{1-z}{2}$. Its variance is given by:

$$\mathrm{Var}(e) = b^2 \left[\frac{\tau}{(1+z)^2} + \frac{1-\tau}{(1-z)^2} - \frac{1}{4}\left(\frac{1-\tau}{1-z} - \frac{\tau}{1+z}\right)^2\right].$$

This expression was used in the simulations to set the scale parameter b to match the target signal-to-noise ratio.

In our simulations, for each fixed value of $\tau \in \{0.25, 0.5, 0.75, 0.95\}$, we set $z = 1 - 2\tau$ to ensure consistency with the theoretical limiting density. The observed process is then simulated as:

$$I(t) = A \exp\left(-\frac{(t-L)^2}{S^2}\right) + e(t), \quad \text{with } e(t) \sim f_{z,b},$$

where we fix $A = 1$, $L = 22$, $S = 10$, and a high signal-to-noise ratio $\mathrm{SNR} = 1000$.

For each combination of $\tau \in \{0.25, 0.5, 0.75, 0.95\}$ and $T \in \{4, 6, 8, 9, 10\}$, we compute the quantile loss profile:

$$f_\tau(l) = \min_{a,s>0} \frac{1}{T} \sum_{t=1}^{T} \rho_\tau \left(I(t) - a \exp\left(-\frac{(t-l)^2}{s^2}\right)\right),$$

where ρ_τ is the asymmetric check loss used in quantile regression.

As illustrated in Figure 4, this construction avoids any need for manual shifting or recentering of the noise, and directly reflects the limiting behavior described in Proposition 2.

5 General Asymmetric Laplace Noise Model

Building on the general density $f_{\tau,z,b}(x)$ introduced in Section 4.1, we now examine the case where $z \neq 0$. Varying the asymmetry parameter z allows us to simulate skewed noise distributions while preserving the property that the τ-quantile lies at zero. This setup enables us to assess how skewness influences the estimation of the peak parameters (a, l, s) under quantile

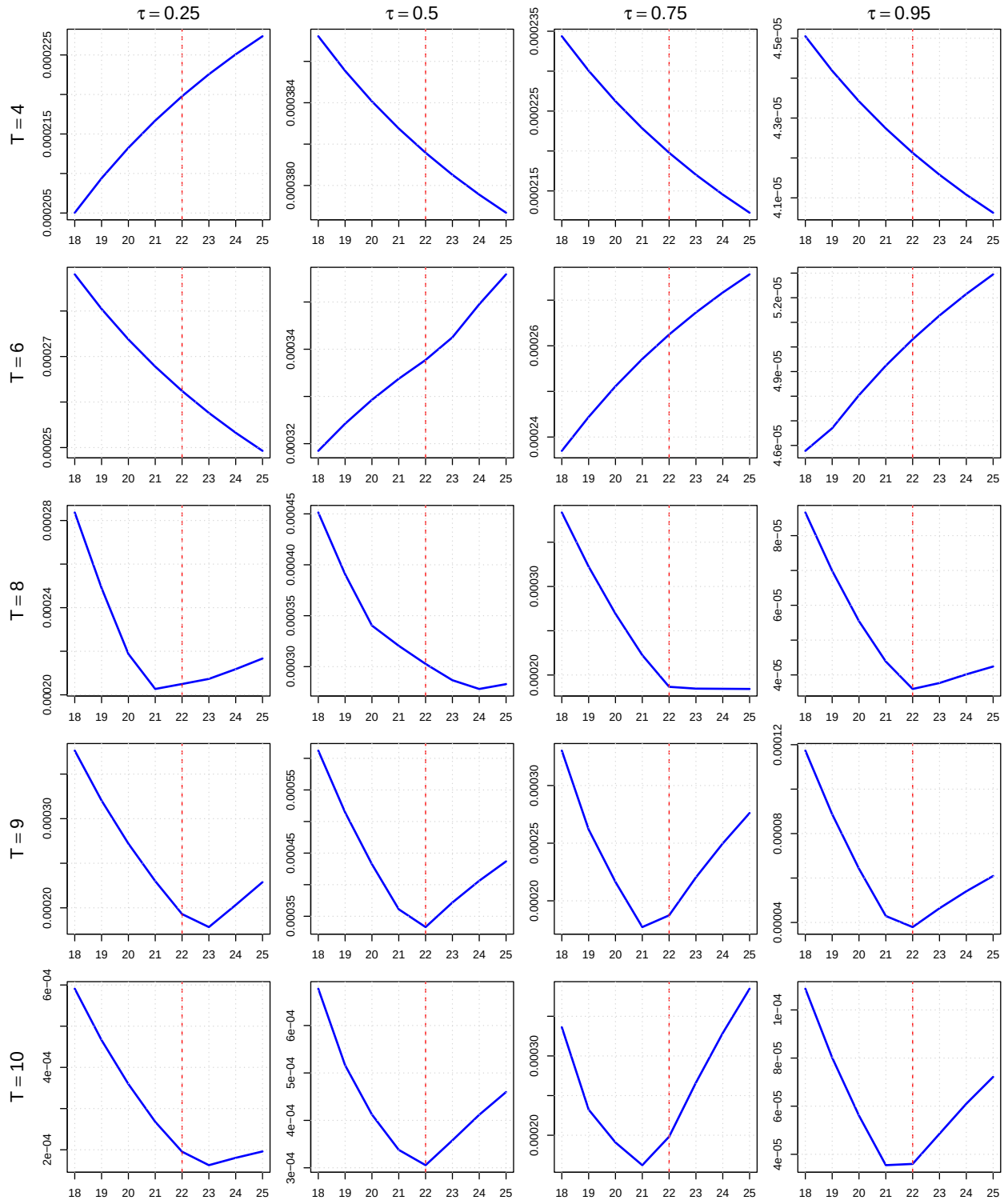


Figure 4: Objective function profiles $l \mapsto f_\tau(T, a, l, s)$ under asymmetric Laplace noise with skewness parameter $z = 1 - 2\tau$, corresponding to the theoretical limiting density derived in Proposition 2. Each panel corresponds to a pair (T, τ), and the red dashed line marks the true peak location $L = 22$.

loss.

5.1 Objective Profiles Across Varying z

We simulate observations according to the model

$$I(t) = A \exp\left(-\frac{(t-L)^2}{S^2}\right) + e(t),$$

where the noise $e(t)$ is drawn from the generalized Laplace distribution $GL_{\tau,z,b}(x)$, whose density is given by:

$$f_{\tau,z,b}(x) = \begin{cases} \dfrac{\tau(1+z)}{b} \exp\left(\dfrac{(1+z)x}{b}\right), & x < 0, \\ \dfrac{(1-\tau)(1-z)}{b} \exp\left(-\dfrac{(1-z)x}{b}\right), & x \geq 0, \end{cases}$$

where $\tau = \frac{1-z}{2}$, ensuring that $\mathbb{P}(e(t) < 0) = \tau$. The scale parameter b is adjusted to ensure constant signal-to-noise ratio using the theoretical variance:

$$\mathrm{Var}(e) = b^2 \left[\frac{\tau}{(1+z)^2} + \frac{1-\tau}{(1-z)^2} - \frac{1}{4}\left(\frac{1-\tau}{1-z} - \frac{\tau}{1+z}\right)^2\right].$$

The signal component is fixed with parameters $A = 15000$, $L = 22$, and $S = 10$, while the scale b is dynamically adjusted to maintain a constant signal-to-noise ratio $\mathrm{SNR} = 1000$ across all configurations. For each combination of quantile level $\tau \in \{0.25, 0.5, 0.75, 0.95\}$ and sample size $T \in \{2, 3, 10, 15, 20, 21\}$, we evaluate the quantile loss objective function

$$f_\tau(l; T) = \min_{a,s>0} \frac{1}{T} \sum_{t=1}^{T} \rho_\tau \left(I(t) - a \exp\left(-\frac{(t-l)^2}{s^2}\right)\right),$$

over a grid $l \in \{18, \ldots, 25\}$.

Figure 5 presents the resulting objective profiles $f_\tau(l; T)$. Each row corresponds to a sample size T, and each column to a quantile level τ. Within each panel, the curves for different values of z are superposed for visual comparison.

To better isolate the effect of skewness for each quantile level, we generate composite plots that fix τ and overlay all z-dependent curves, shown in Figure 6.

These overlays allow direct comparison of how different noise asymmetries influence the curvature, location, and convexity of the objective function for a given estimation target.

6 Parameter Estimation Trajectories

We now analyze how the estimated model parameters, location $\hat{l}(T)$, amplitude $\hat{a}(T)$, and scale $\hat{s}(T)$, evolve as the sample size T increases. Our goal is to assess the stability and convergence of these estimators under both symmetric and asymmetric Laplace noise. All experiments use a

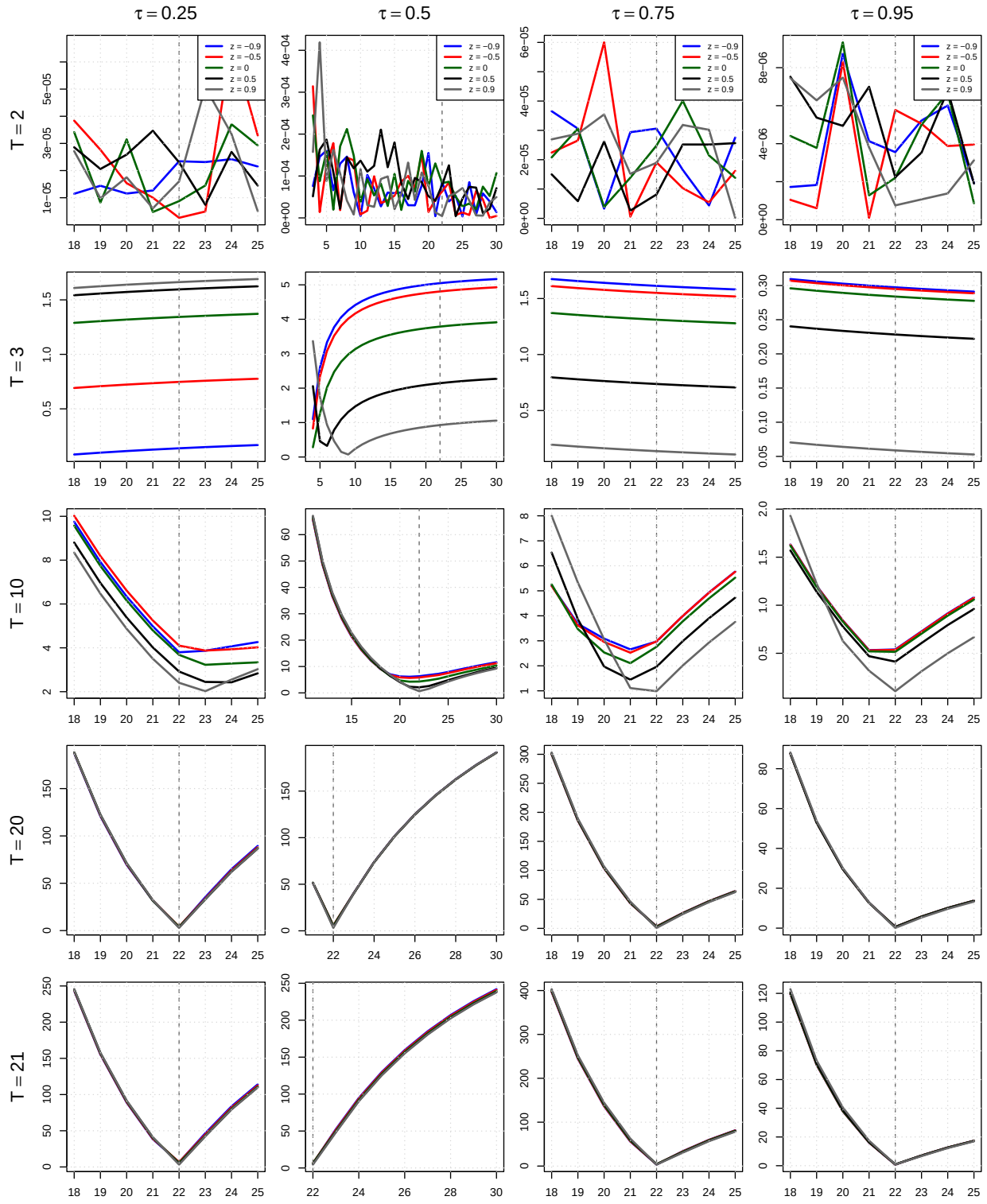


Figure 5: Quantile loss objective $l \mapsto f_\tau(l;T)$ under asymmetric Laplace noise with skewness parameters $z \in \{-0.9, -0.5, 0, 0.5, 0.9\}$, derived from the limiting form $x \mapsto \exp\left(-\frac{|x|}{b}(1-\operatorname{sgn}(x)z)\right)$. Rows correspond to sample sizes $T \in \{2, 3, 10, 20, 21\}$, and columns correspond to quantile levels $\tau \in \{0.25, 0.5, 0.75, 0.95\}$. The vertical dashed line marks the true peak location $L = 22$. Colors indicate the value of z.

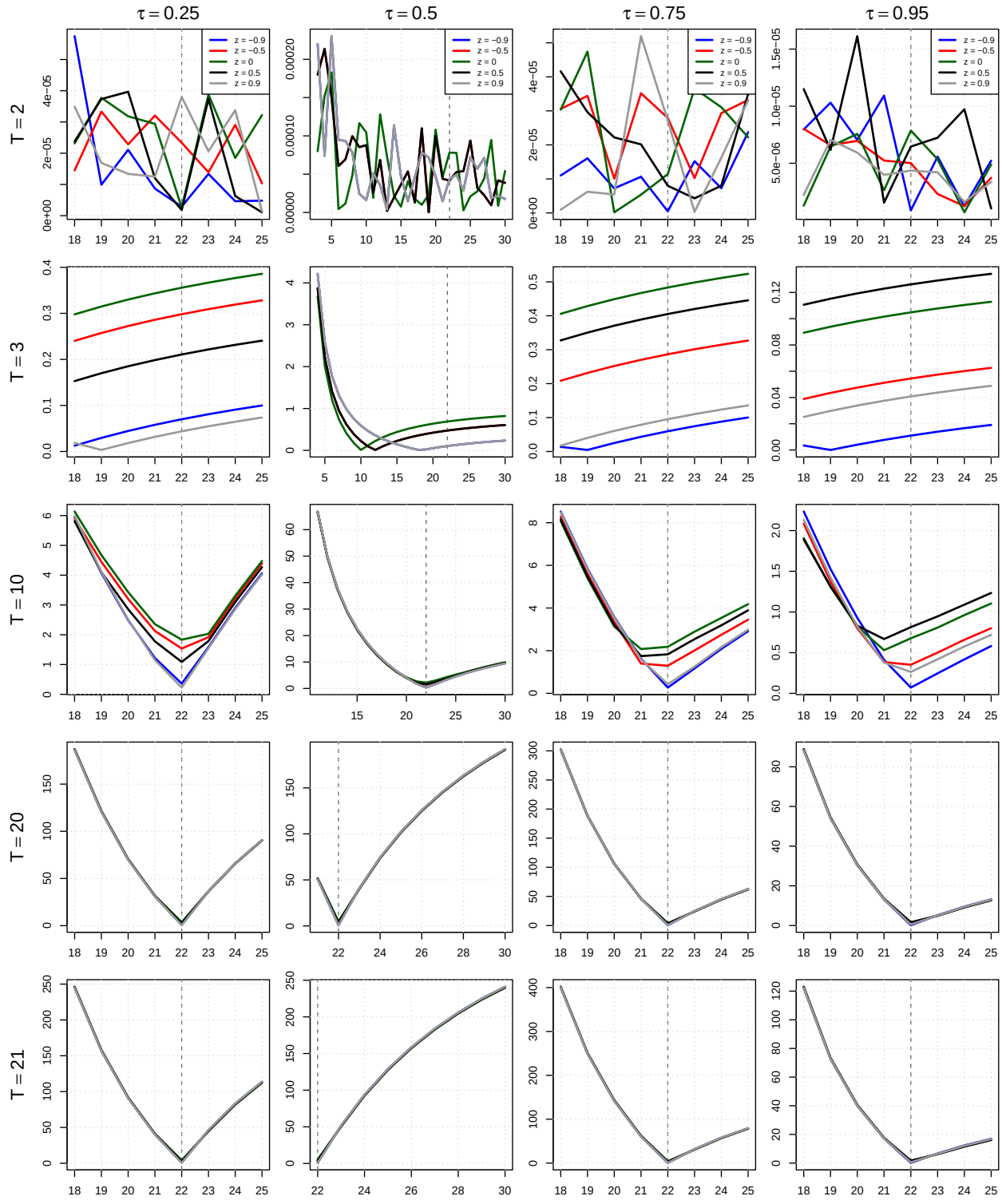


Figure 6: Superposed objective functions $l \mapsto f_\tau(l;T)$ under asymmetric Laplace noise with varying skewness parameters $z \in \{-0.9, -0.5, 0, 0.5, 0.9\}$. Each row corresponds to sample sizes $T \in \{2, 3, 10, 20, 21\}$, and each column to quantile levels $\tau \in \{0.25, 0.5, 0.75, 0.95\}$. The vertical dashed line indicates the true peak location $L = 22$; colors denote different values of z.

fixed signal-to-noise ratio of SNR $= 1000$ and consider quantile levels $\tau \in \{0.25, 0.5, 0.75, 0.95\}$.

6.1 Symmetric Noise (z = 0)

We first consider the case of symmetric Laplace noise, corresponding to $z = 0$. The observations are generated according to the signal-plus-noise model

$$I(t) = A \exp\left(-\frac{(t-L)^2}{S^2}\right) + e(t), \quad t = 1, \ldots, 30,$$

with $A = 1$, $L = 22$, and $S = 10$. The noise $e(t)$ follows the generalized Laplace distribution $GL_{\tau,b}(u)$, with density proportional to $\exp(-\rho_\tau(u/b))$, where $\rho_\tau(u) = u(\tau - \mathbf{1}_{\{u<0\}})$. The scale b is calibrated so that the noise variance matches $(A/\text{SNR})^2$, using the expression:

$$\text{Var}(e) = b^2 \left[\frac{\tau}{(1+z)^2} + \frac{1-\tau}{(1-z)^2} - \frac{1}{4}\left(\frac{1-\tau}{1-z} - \frac{\tau}{1+z}\right)^2\right],$$

with $z = 0$ for the symmetric case.

At each time horizon $T \in [4, 30]$, we estimate the parameters $(\hat{l}(T), \hat{a}(T), \hat{s}(T))$ by minimizing the quantile loss:

$$\min_{l \in [1,h], s>0} \frac{1}{T} \sum_{t=1}^{T} \rho_\tau \left(I(t) - a \exp\left(-\frac{(t-l)^2}{s^2}\right)\right),$$

where $\rho_\tau(u) = u(\tau - \mathbf{1}_{\{u<0\}})$ is the check function, and a is re-estimated for each (l, s) via a weighted τ-quantile.

Figure 7 shows the estimation trajectories $T \mapsto \hat{l}(T)$, $T \mapsto \hat{a}(T)$, and $T \mapsto \hat{s}(T)$ for each quantile level.

6.2 Asymmetric Noise with Varying Skewness Parameter z

We investigate the influence of asymmetry in the additive noise distribution on the estimation trajectories. The signal retains the same Gaussian-shaped form as previously:

$$I(t) = A \exp\left(-\frac{(t-L)^2}{S^2}\right) + e(t), \quad t = 1, \ldots, T,$$

where the noise $e(t)$ is generated from the generalized Laplace distribution

$$f_{\tau,b}(x) \propto \exp\left(-\rho_\tau\left(\frac{x}{b}\right)\right),$$

with asymmetry parameter $z \in \{-0.9, -0.5, 0, 0.5, 0.9\}$, quantile level $\tau \in (0, 1)$, and scale $b > 0$.

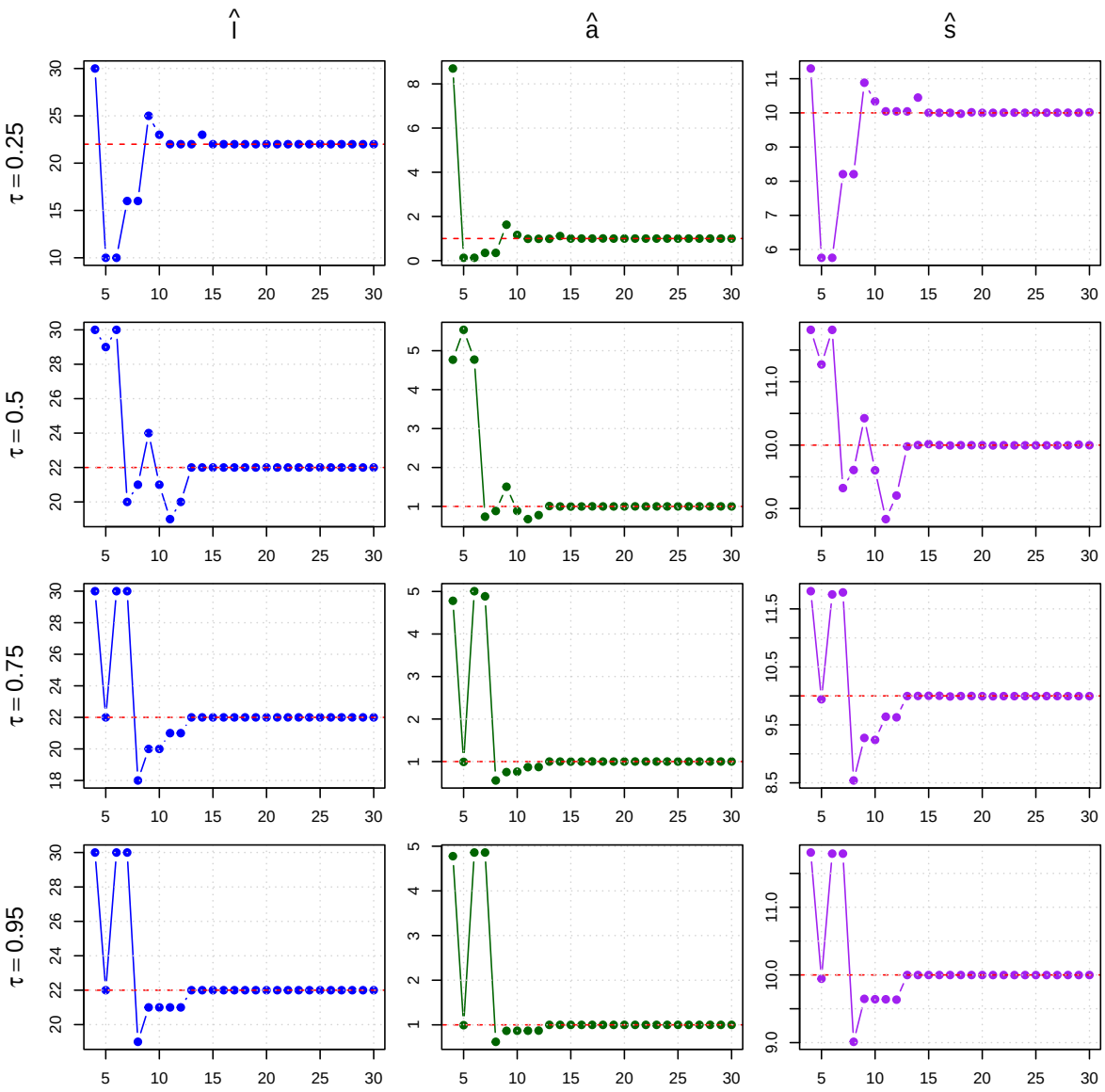


Figure 7: Estimation paths $T \mapsto \hat{l}(T)$ (left), $\hat{a}(T)$ (middle), and $\hat{s}(T)$ (right) under symmetric Laplace noise with signal-to-noise ratio (SNR) 1000. Each row corresponds to a quantile level $\tau \in \{0.25, 0.5, 0.75, 0.95\}$. The true values $(L, A, S) = (22, 1, 10)$ are indicated by red dashed lines.

The skewness and quantile level are linked via:

$$P(e_t < 0) = \tau, \quad \text{with} \quad \tau = \frac{1-z}{2},$$

and the scale b is calibrated such that the noise variance satisfies the fixed signal-to-noise ratio (SNR) condition:

$$\mathrm{Var}(e) = \left(\frac{A}{\mathrm{SNR}}\right)^2.$$

Using the exact variance of the asymmetric Laplace distribution $f_{\tau,b}$, the scale is computed as:

$$b = \sqrt{\frac{(A/\mathrm{SNR})^2}{\frac{\tau}{(1+z)^2} + \frac{1-\tau}{(1-z)^2} - \frac{1}{4}\left(\frac{1-\tau}{1-z} - \frac{\tau}{1+z}\right)^2}}.$$

The estimation procedure remains unchanged and is applied sequentially for increasing sample sizes $T = 4, \ldots, 30$.

Figure 8 displays the estimation paths for the parameters $\hat{l}(T)$, $\hat{a}(T)$, and $\hat{s}(T)$ for each combination of τ and z, with true parameter values $(L, A, S) = (22, 1, 10)$ indicated by dashed horizontal lines.

To ensure comparability, we reuse the variance calibration method described above. Figure 9 shows the resulting trajectories, where each row corresponds to a quantile level and each column to one of the estimated parameters.

7 Stability Analysis

This section investigates the temporal stability of peak estimators obtained by minimizing different loss functions. We begin by analyzing the performance of the quantile loss criterion under high noise scenarios, tracking the evolution of the estimated peak location as new data points become available. We then complement this analysis by introducing a likelihood-based loss derived from the asymmetric Laplace distribution. This second approach enables a probabilistic assessment of stability by incorporating distributional assumptions on the residuals. Together, these perspectives provide a robust evaluation of how estimation paths behave under varying levels of noise and asymmetry.

7.1 Low Signal-to-Noise Performance

To evaluate the robustness of the quantile-based estimator under high noise, we simulate estimation paths under a signal-to-noise ratio of SNR $= 40$. We vary both the quantile level $\tau \in \{0.25, 0.5, 0.75, 0.95\}$ and the asymmetry parameter $z \in \{-0.9, -0.5, 0, 0.5, 0.9\}$. For each configuration, we compute the path $T \mapsto \hat{l}(T)$ as the number of observations T increases from 4 to 35. Entry and exit times into a stable region are detected using thresholds $\varepsilon = 4$ and $\varepsilon' = 0$, respectively.

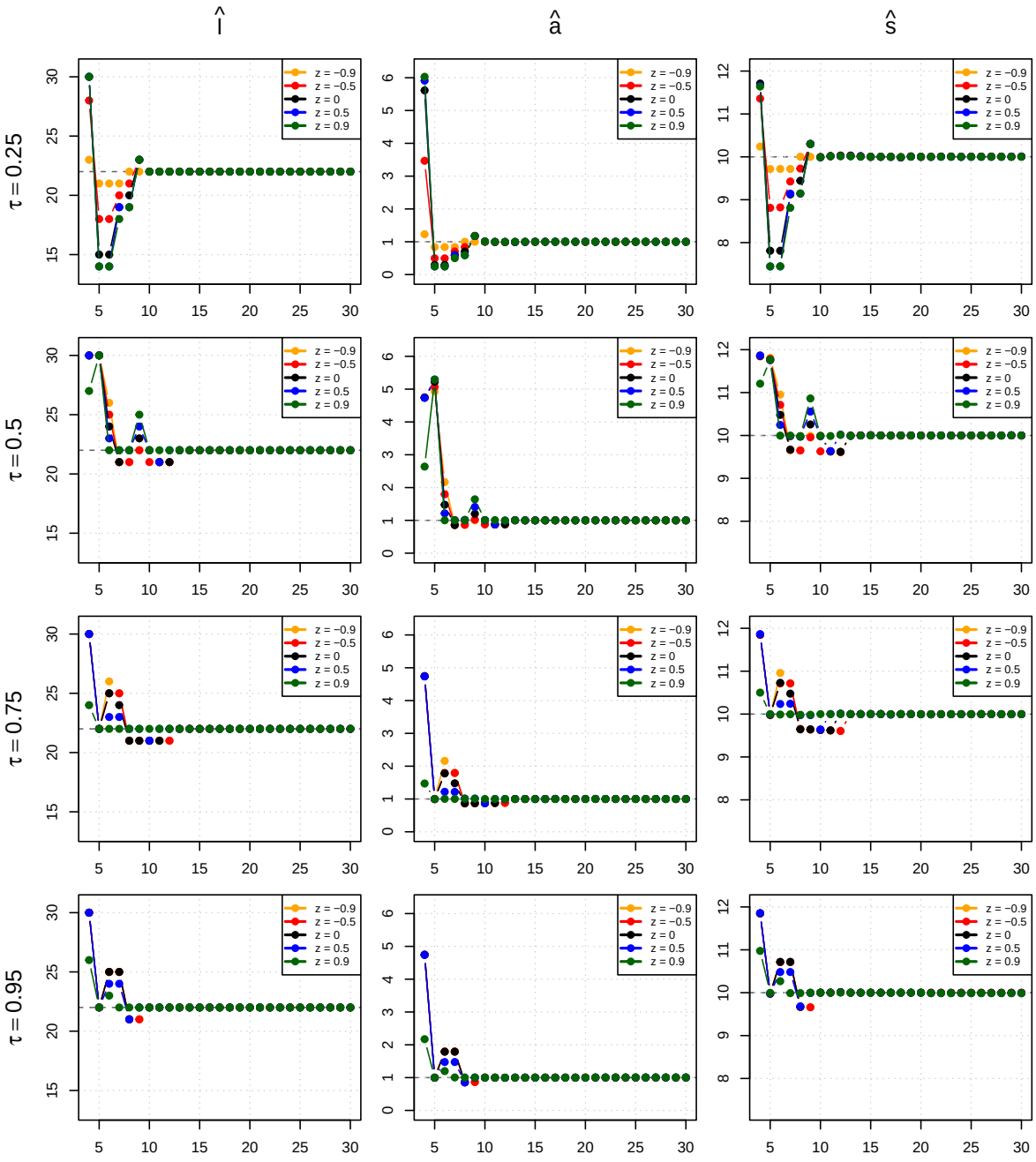


Figure 8: Estimation trajectories $T \mapsto \hat{l}(T)$ (left column), $\hat{a}(T)$ (middle column), and $\hat{s}(T)$ (right column) under asymmetric Laplace noise generated from $f_{\tau,b}(x) \propto \exp(-\rho_\tau(x/b))$, with skewness $z \in \{-0.9, -0.5, 0, 0.5, 0.9\}$ and fixed signal-to-noise ratio SNR $= 1000$. Each row corresponds to a quantile level $\tau \in \{0.25, 0.5, 0.75, 0.95\}$. Colored lines track estimates as the sample size increases from $T = 4$ to $T = 30$. True parameter values $(L, A, S) = (22, 1, 10)$ are shown as horizontal dashed lines for reference.

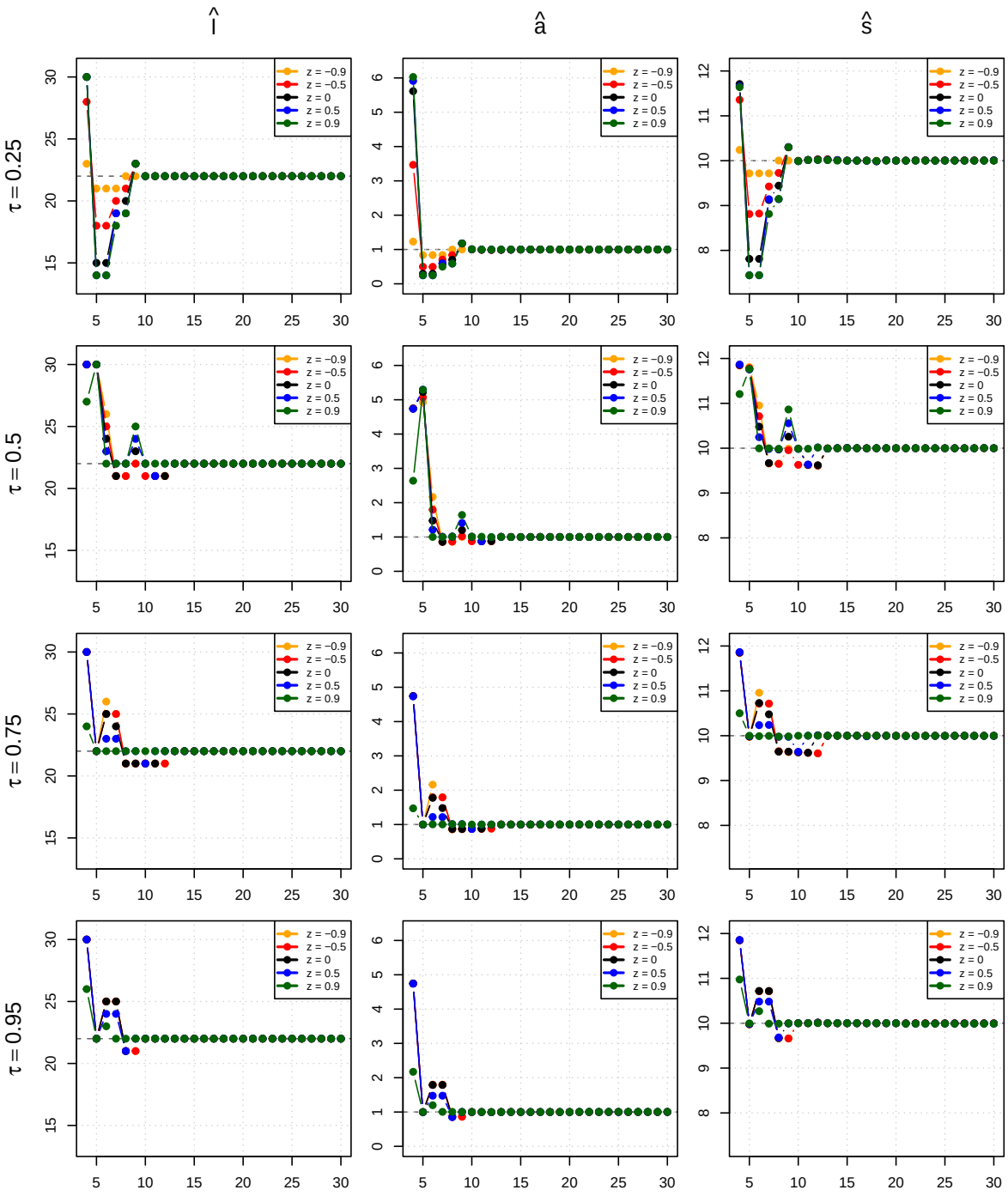


Figure 9: Estimation paths $T \mapsto \hat{l}(T)$, $\hat{a}(T)$, and $\hat{s}(T)$ under a general asymmetric Laplace noise model with skewness parameter $z \in \{-0.9, -0.5, 0, 0.5, 0.9\}$ and fixed signal-to-noise ratio $\mathrm{SNR} = 1000$. Rows correspond to quantile levels $\tau \in \{0.25, 0.5, 0.75, 0.95\}$, columns correspond to estimated parameters. True parameter values $(L, A, S) = (22, 1, 10)$ are indicated by dashed horizontal lines.

Figure 10 shows the estimation paths $\hat{l}(T)$ for each τ and z, with the true peak location $L = 22$ marked by a horizontal dashed line. The corresponding entry and exit times are reported in Table 1.

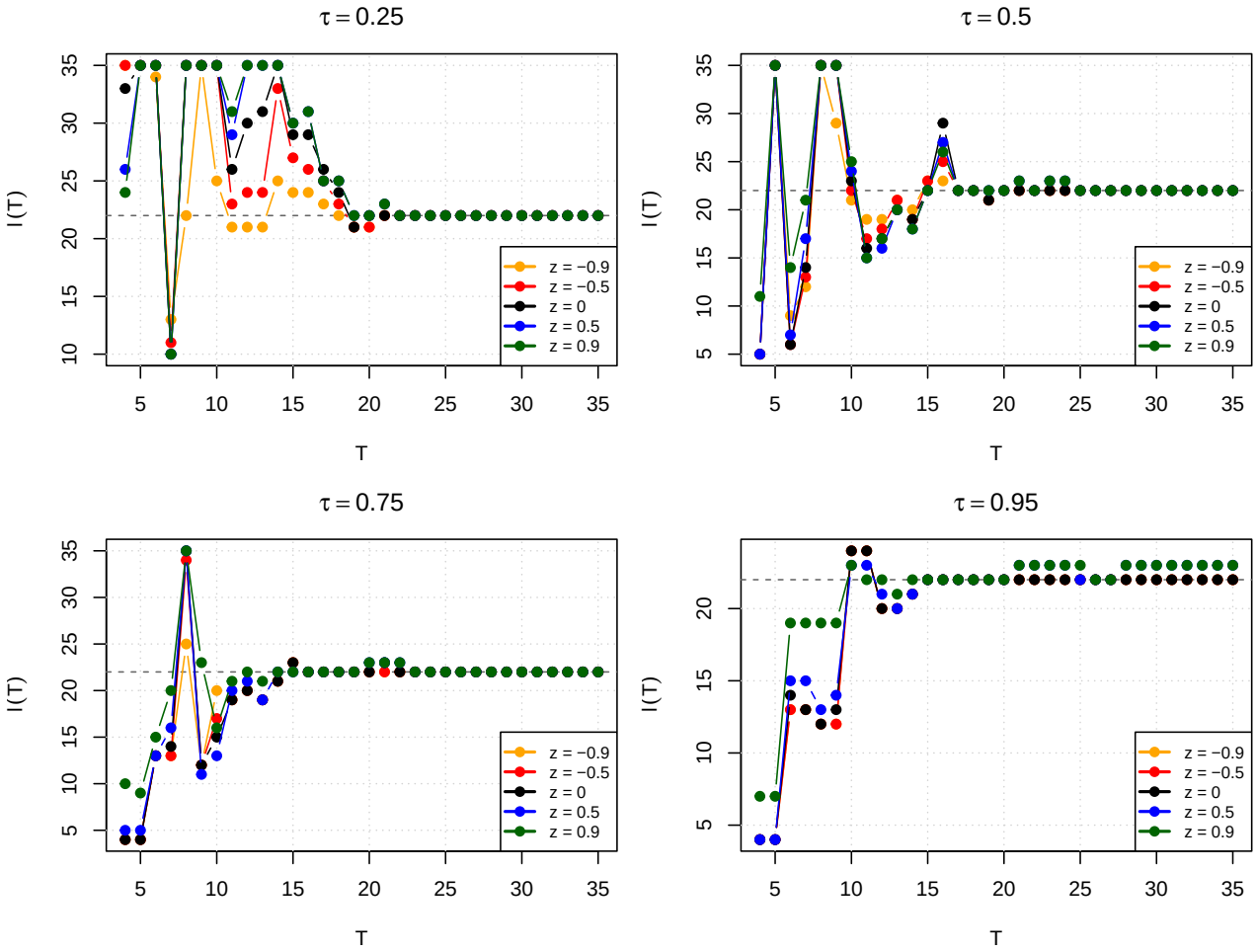


Figure 10: Estimation paths $T \mapsto \hat{l}(T)$ under low signal-to-noise ratio (SNR = 40), for quantile levels $\tau \in \{0.25, 0.5, 0.75, 0.95\}$. Each panel corresponds to a different value of τ, and each curve within a panel represents a different asymmetry parameter $z \in \{-0.9, -0.5, 0, 0.5, 0.9\}$. The horizontal dashed line indicates the true peak location $L = 22$.

Table 1: Entry (T_{entry}) and exit (T_{exit}) times for each (τ, z) combination under SNR = 40.

z	$T_{\text{entry}}^{\tau=0.25}$	$T_{\text{entry}}^{\tau=0.5}$	$T_{\text{entry}}^{\tau=0.75}$	$T_{\text{entry}}^{\tau=0.95}$	$T_{\text{exit}}^{\tau=0.25}$	$T_{\text{exit}}^{\tau=0.5}$	$T_{\text{exit}}^{\tau=0.75}$	$T_{\text{exit}}^{\tau=0.95}$
-0.9	10.0	10.0	10.0	10.0	21.0	20.0	16.0	15.0
-0.5	17.0	15.0	11.0	10.0	21.0	20.0	16.0	15.0
0.0	18.0	17.0	11.0	10.0	20.0	20.0	22.0	15.0
0.5	17.0	17.0	11.0	10.0	22.0	25.0	23.0	28.0
0.9	17.0	15.0	11.0	6.0	22.0	25.0	23.0	28.0

7.2 Stability Analysis Based on Log-Likelihood Loss

To complement our quantile loss analysis, we now evaluate stability using a log-likelihood-based criterion derived from the asymmetric Laplace density introduced in Section 4.1. Specifically, we consider the negative log-likelihood objective:

$$\mathcal{L}_{\tau,z,b}(I; a, l, s) = -\sum_{t=1}^{T} \log f_{\tau,z,b}\left(I(t) - a\exp\left(-\frac{(t-l)^2}{s^2}\right)\right),$$

where $f_{\tau,z,b}$ denotes the asymmetric Laplace density parameterized by quantile level τ, asymmetry z, and scale b.

This likelihood-based loss function directly incorporates the assumed noise distribution, offering a probabilistically motivated alternative to the piecewise linear quantile loss. Minimizing $\mathcal{L}_{\tau,z,b}$ over (a, l, s) provides a maximum likelihood estimate of the parameters under the asymmetric Laplace error model.

As in the previous section, we use the resulting estimator $\hat{l}(T)$ to track the estimated peak location over time. Entry time T_{entry} is defined as the earliest time at which the estimation path stabilizes within a fixed window, and exit time T_{exit} marks the point at which it departs from that window. We apply this procedure across multiple values of $\tau \in \{0.25, 0.5, 0.75, 0.95\}$ and asymmetry levels $z \in \{-0.9, -0.5, 0, 0.5, 0.9\}$, and report the results under two regimes: 1) Moderate Noise: Signal-to-noise ratio (SNR) fixed at 1000, 2) High Noise: SNR reduced to 40.

Table 2: Entry and exit times based on log-likelihood loss, for each (τ, z) under SNR = 1000.

z	$T_{\text{entry}}^{\tau=0.25}$	$T_{\text{entry}}^{\tau=0.5}$	$T_{\text{entry}}^{\tau=0.75}$	$T_{\text{entry}}^{\tau=0.95}$	$T_{\text{exit}}^{\tau=0.25}$	$T_{\text{exit}}^{\tau=0.5}$	$T_{\text{exit}}^{\tau=0.75}$	$T_{\text{exit}}^{\tau=0.95}$
-0.9	10.0	9.0	10.0	10.0	20.0	15.0	14.0	14.0
-0.5	11.0	10.0	10.0	10.0	18.0	17.0	14.0	14.0
0.0	15.0	13.0	10.0	10.0	19.0	17.0	14.0	14.0
0.5	15.0	13.0	11.0	10.0	19.0	17.0	14.0	14.0
0.9	15.0	13.0	9.0	7.0	19.0	17.0	11.0	22.0

Table 3: Entry and exit times based on log-likelihood loss, for each (τ, z) under SNR = 40.

z	$T_{\text{entry}}^{\tau=0.25}$	$T_{\text{entry}}^{\tau=0.5}$	$T_{\text{entry}}^{\tau=0.75}$	$T_{\text{entry}}^{\tau=0.95}$	$T_{\text{exit}}^{\tau=0.25}$	$T_{\text{exit}}^{\tau=0.5}$	$T_{\text{exit}}^{\tau=0.75}$	$T_{\text{exit}}^{\tau=0.95}$
-0.9	10.0	10.0	10.0	10.0	21.0	20.0	16.0	15.0
-0.5	17.0	15.0	11.0	10.0	21.0	20.0	16.0	15.0
0.0	18.0	17.0	11.0	10.0	20.0	20.0	22.0	15.0
0.5	17.0	17.0	11.0	10.0	22.0	25.0	23.0	28.0
0.9	17.0	15.0	11.0	6.0	22.0	25.0	23.0	28.0

8 Real-World Application: COVID-19 Case Study

We now demonstrate the application of our quantile-based peak estimation framework on real-world data. We focus on the daily confirmed COVID-19 cases in China, France, and Germany. Figure 11 displays the daily case series for these three countries. For each country, we analyze the sequence $T \mapsto \hat{l}(T)$ for various quantile levels $\tau \in \{0.25, 0.5, 0.75, 0.95\}$, using quantile loss minimization over a Gaussian-shaped parametric model (parametrized only by τ). We identify the entry and exit times into the region where $\hat{l}(T)$ stabilizes, based on thresholds ε and ε' specific to each country, and the parameter τ.

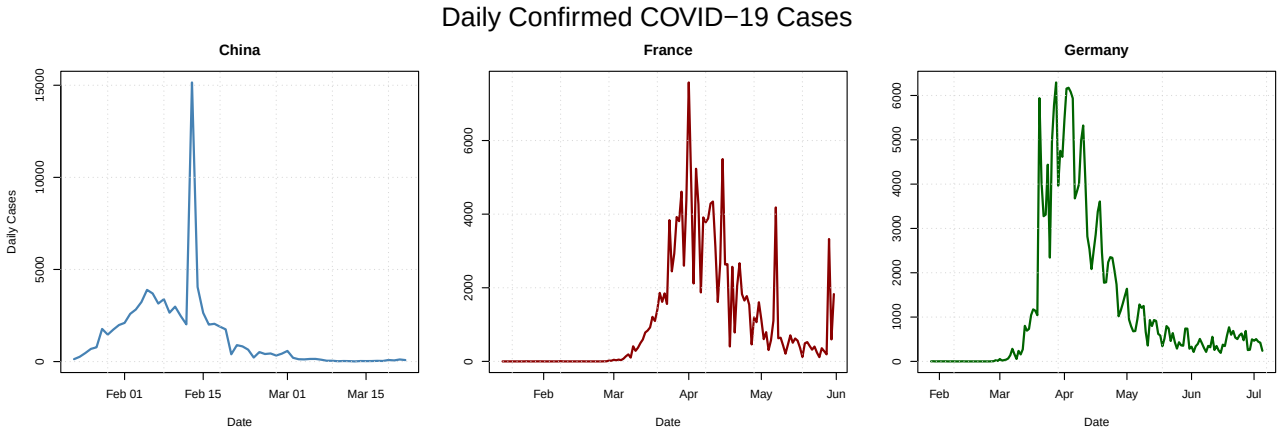


Figure 11: Daily confirmed COVID-19 cases in China, France, and Germany (left to right).

8.1 China

We analyze the first wave of the COVID-19 epidemic in China, using the first 60 daily new case counts from January 22 to March 22, 2020. The smoothing parameter is set to $h = 35$, and the thresholds are $\varepsilon = 3$ and $\varepsilon' = 0$. The results for the four quantile levels are reported below. The estimated peak location sequences for $\tau = 0.25$, 0.5, 0.75, and 0.95 are shown in Table 4.

Table 4: Estimated peak location $\hat{l}(T)$ for COVID-19 in China using quantile level $\tau \in \{0.25, 0.5, 0.75, 0.95\}$

$\tau = 0.25, T_{entry} = 15, T_{exit} = 27, \varepsilon = 3, \varepsilon' = 0$																
T	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
$\hat{l}(T)$	6	5	35	14	11	11	11	11	14	14	21	18	15	15	15	15
T	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35
$\hat{l}(T)$	15	15	15	15	15	15	15	16	16	16	16	16	16	16	16	16
$\tau = 0.5, T_{entry} = 13, T_{exit} = 30, \varepsilon = 3, \varepsilon' = 0$																
T	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
$\hat{l}(T)$	6	5	35	8	9	11	11	12	14	15	18	17	16	16	15	16
T	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35
$\hat{l}(T)$	15	15	15	16	16	17	17	18	18	18	17	17	17	17	17	17
$\tau = 0.75, T_{entry} = 23, T_{exit} = 26, \varepsilon = 3, \varepsilon' = 0$																
T	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
$\hat{l}(T)$	6	5	35	8	7	8	10	13	14	16	24	18	17	16	15	15
T	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35
$\hat{l}(T)$	15	15	35	20	20	20	18	18	18	18	18	18	18	18	18	18
$\tau = 0.95, T_{entry} = 28, T_{exit} = 31, \varepsilon = 3, \varepsilon' = 0$																
T	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
$\hat{l}(T)$	6	5	35	9	7	8	9	10	11	15	29	29	15	14	14	15
T	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35
$\hat{l}(T)$	15	14	35	35	35	35	35	26	22	21	21	20	20	20	20	20

8.2 France

Next, we apply the estimation method to daily COVID-19 cases in France, using ECDC data from January 15 to May 31, 2020. The analysis covers $T \in [50, 89]$, with a smoothing window of $h = 120$. We use thresholds $\varepsilon = 5$ and $\varepsilon' = 1$ to identify the stable peak zone. Results for the four values of τ are presented below. The estimated peak location sequences for $\tau = 0.25$, 0.5, 0.75, and 0.95 are reported in Table 5.

Table 5: Estimated peak location $\hat{l}(T)$ for COVID-19 in France using quantile level $\tau \in \{0.25, 0.5, 0.75, 0.95\}$

$\tau = 0.25, T_{entry} = 85, T_{exit} = 86, \varepsilon = 5, \varepsilon' = 1$																				
T	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69
$\hat{l}(T)$	63	61	120	120	56	119	96	71	83	71	72	67	65	66	66	66	74	72	72	69
T	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89
$\hat{l}(T)$	69	72	81	82	90	92	76	80	89	88	75	79	83	76	76	80	82	82	82	82
$\tau = 0.5, T_{entry} = 82, T_{exit} = 86, \varepsilon = 5, \varepsilon' = 1$																				
T	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69
$\hat{l}(T)$	49	54	88	72	54	120	58	60	120	87	108	71	69	77	67	72	73	71	73	70
T	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89
$\hat{l}(T)$	74	78	78	83	88	90	88	85	90	86	85	86	83	80	79	79	80	80	81	80
$\tau = 0.75, T_{entry} = 80, T_{exit} = 87, \varepsilon = 5, \varepsilon' = 1$																				
T	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69
$\hat{l}(T)$	48	51	120	120	120	120	120	59	64	65	87	68	66	81	73	70	83	83	72	72
T	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89
$\hat{l}(T)$	120	89	78	84	84	86	85	83	120	87	83	81	81	81	80	79	79	81	81	81
$\tau = 0.95, T_{entry} = 85, T_{exit} = 87, \varepsilon = 5, \varepsilon' = 1$																				
T	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69
$\hat{l}(T)$	48	51	56	106	59	120	120	69	59	60	63	63	64	67	67	70	94	87	77	70
T	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89
$\hat{l}(T)$	120	120	120	120	79	76	76	76	89	89	89	86	86	86	85	83	80	78	79	79

8.3 Germany

Finally, we consider COVID-19 daily cases in Germany from January 28 to July 5, 2020. To align the peak location with day $t = 61$, we index the data starting on January 28. We analyze $T \in [40, 75]$ with a smoothing window of $h = 160$. Due to the more complex shape of the infection curve, we adopt larger thresholds: $\varepsilon = 15$ and $\varepsilon' = 2$. The peak estimation results for the four quantile levels are summarized below. Table 6 displays the estimation results for $\tau = 0.25$, 0.5, 0.75, and 0.95.

Table 6: Estimated peak location $\hat{l}(T)$ for COVID-19 in Germany using quantile level $\tau \in \{0.25, 0.5, 0.75, 0.95\}$

$\tau = 0.25, T_{entry} = 61, T_{exit} = 70, \varepsilon = 15, \varepsilon' = 2$																		
T	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57
$\hat{l}(T)$	160	45	42	41	44	47	160	88	48	160	54	52	51	51	110	160	96	96
T	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
$\hat{l}(T)$	56	56	78	71	60	61	62	62	68	75	72	70	65	65	65	66	66	66
$\tau = 0.5, T_{entry} = 59, T_{exit} = 67, \varepsilon = 15, \varepsilon' = 2$																		
T	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57
$\hat{l}(T)$	159	40	40	46	42	46	160	87	64	74	68	58	51	160	160	102	73	72
T	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
$\hat{l}(T)$	56	63	74	71	62	61	61	61	63	65	66	66	66	64	64	65	66	66
$\tau = 0.75, T_{entry} = 56, T_{exit} = 66, \varepsilon = 15, \varepsilon' = 2$																		
T	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57
$\hat{l}(T)$	160	40	40	44	42	47	160	144	54	106	60	57	52	160	160	160	55	56
T	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
$\hat{l}(T)$	56	61	66	67	66	64	60	62	64	64	65	65	65	65	64	65	66	66
$\tau = 0.95, T_{entry} = 57, T_{exit} = 73, \varepsilon = 15, \varepsilon' = 2$																		
T	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57
$\hat{l}(T)$	160	50	43	41	41	43	53	53	53	53	54	54	54	160	160	160	71	55
T	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
$\hat{l}(T)$	53	55	57	58	58	57	58	59	60	61	61	61	61	61	61	64	66	66

8.4 Comparison Across Quantile Levels

To compare the performance of different quantile levels, we compute the value of the quantile loss-based objective function at the final time point $T_{\max}$ for each country. This allows us to identify the quantile level τ that best fits the observed trajectory. The best-performing quantiles (i.e., the ones minimizing the objective function at $T_{\max}$) are:

- China: $\tau = 0.25$ at $T = 35$,
- France: $\tau = 0.95$ at $T = 89$,
- Germany: $\tau = 0.95$ at $T = 75$.

These values indicate that different regions of the conditional distribution may best capture the epidemic dynamics in different countries, with higher quantile levels offering better fits for France and Germany, while lower quantiles provide better performance for China.

8.5 Stability Analysis Using Asymmetric Laplace Log-Likelihood

To further investigate the robustness of our peak location estimates, we extend the stability analysis introduced in Section 7 to real COVID-19 case data. Specifically, we replace the quantile loss objective by the asymmetric Laplace-based objective

$$-\log(f_{\tau,z,b}(e)) = -\log\left(\frac{\tau(1-z)+(1-\tau)(1+z)}{2b}\right) + \frac{|e|}{b}\left[(1+z)\mathbf{1}_{\{e<0\}} + (1-z)\mathbf{1}_{\{e\geq 0\}}\right],$$

where $e = I(t) - a \cdot \exp\left(-\frac{(t-l)^2}{s^2}\right)$, and (τ, z, b) parameterize the asymmetric Laplace distribution.

Using the real daily case data for China, France, and Germany, we estimate the entry and exit times into the stable region of the estimated peak location path $T \mapsto \hat{l}(T)$ across combinations of quantile levels $\tau \in \{0.25, 0.5, 0.75, 0.95\}$ and asymmetry parameters $z \in \{-0.9, -0.5, 0, 0.5, 0.9\}$. The scale parameter is fixed to $b = 1$, and thresholds ε, ε' are set to country-specific values determined by the complexity of each epidemic curve.

The following tables summarize the entry and exit times computed from the log-likelihood loss for each country:

Table 7: Entry and exit times based on $-\log(f_{\tau,z,b}(e))$ objective, for each (τ, z) for China, using thresholds $\varepsilon = 3$ and $\varepsilon' = 0$.

z	$T_{\text{entry}}^{\tau=0.25}$	$T_{\text{entry}}^{\tau=0.5}$	$T_{\text{entry}}^{\tau=0.75}$	$T_{\text{entry}}^{\tau=0.95}$	$T_{\text{exit}}^{\tau=0.25}$	$T_{\text{exit}}^{\tau=0.5}$	$T_{\text{exit}}^{\tau=0.75}$	$T_{\text{exit}}^{\tau=0.95}$
-0.9	29.0	19.0	19.0	25.0	32.0	33.0	30.0	33.0
-0.5	29.0	19.0	14.0	24.0	35.0	33.0	27.0	27.0
0.0	24.0	22.0	14.0	13.0	30.0	31.0	30.0	28.0
0.5	29.0	17.0	13.0	13.0	33.0	33.0	34.0	19.0
0.9	29.0	14.0	13.0	21.0	35.0	24.0	16.0	21.0

Table 8: Entry and exit times based on $-\log(f_{\tau,z,b}(e))$ objective, for each (τ, z) for France, using thresholds $\varepsilon = 5$ and $\varepsilon' = 1$.

z	$T_{\text{entry}}^{\tau=0.25}$	$T_{\text{entry}}^{\tau=0.5}$	$T_{\text{entry}}^{\tau=0.75}$	$T_{\text{entry}}^{\tau=0.95}$	$T_{\text{exit}}^{\tau=0.25}$	$T_{\text{exit}}^{\tau=0.5}$	$T_{\text{exit}}^{\tau=0.75}$	$T_{\text{exit}}^{\tau=0.95}$
-0.9	87.0	81.0	77.0	78.0	88.0	87.0	84.0	86.0
-0.5	87.0	81.0	74.0	85.0	88.0	87.0	81.0	86.0
0.0	88.0	81.0	73.0	87.0	89.0	86.0	86.0	87.0
0.5	80.0	84.0	74.0	84.0	88.0	85.0	84.0	87.0
0.9	77.0	85.0	84.0	86.0	86.0	85.0	84.0	86.0

Table 9: Entry and exit times based on $-\log(f_{\tau,z,b}(e))$ objective, for each (τ, z) for Germany, using thresholds $\varepsilon = 15$ and $\varepsilon' = 2$.

z	$T_{\text{entry}}^{\tau=0.25}$	$T_{\text{entry}}^{\tau=0.5}$	$T_{\text{entry}}^{\tau=0.75}$	$T_{\text{entry}}^{\tau=0.95}$	$T_{\text{exit}}^{\tau=0.25}$	$T_{\text{exit}}^{\tau=0.5}$	$T_{\text{exit}}^{\tau=0.75}$	$T_{\text{exit}}^{\tau=0.95}$
-0.9	54.0	51.0	51.0	50.0	70.0	69.0	68.0	67.0
-0.5	55.0	53.0	51.0	50.0	73.0	72.0	67.0	66.0
0.0	55.0	51.0	53.0	53.0	74.0	68.0	73.0	61.0
0.5	54.0	55.0	54.0	53.0	69.0	70.0	71.0	72.0
0.9	53.0	55.0	75.0	75.0	63.0	71.0	75.0	75.0

8.6 Entry and Exit Times Using the GAL Loss

We now investigate the stability of the estimated peak location using an alternative loss function derived from the Generalized Asymmetric Laplace (GAL) distribution. Specifically, we consider the loss function

$$u \mapsto \frac{\left(z - \frac{\lambda u}{b}\right)^2}{2\lambda} + \frac{|u|}{b},$$

where $z = 1 - 2\tau$ and $\lambda > 0$ is a tuning parameter. This formulation generalizes the standard asymmetric Laplace-based log-likelihood used in our earlier peak estimation procedure, and as $\lambda \to 0$, it converges to a scaled version of the Laplace loss $\rho_\tau(u)/b$, as formalized in Proposition 2.

This robust loss penalizes deviations asymmetrically and allows for heavier or lighter tails depending on the value of λ, providing additional flexibility in modeling the noise structure in real-world epidemiological data.

For each country (China, France, and Germany), we use the same trajectory of daily confirmed cases as in previous sections, and apply the modified estimation procedure based on the GAL loss. For each quantile level $\tau \in \{0.25, 0.5, 0.75, 0.95\}$, we evaluate three values of $\lambda \in \{1, 0.5, 0.1\}$, which control the departure from the asymmetric Laplace limit. Entry and exit times are estimated by analyzing the stability of the estimated peak location $\hat{\ell}(T)$, as in our earlier definition, using thresholds adapted to each country's data characteristics.

Tables 10–12 report the estimated entry and exit times for China, France, and Germany respectively.

Table 10: Entry and exit times based on GAL loss, for each (τ, λ) for China, using thresholds $\varepsilon = 3$ and $\varepsilon' = 0$.

λ	$T_{\text{entry}}^{\tau=0.25}$	$T_{\text{entry}}^{\tau=0.5}$	$T_{\text{entry}}^{\tau=0.75}$	$T_{\text{entry}}^{\tau=0.95}$	$T_{\text{exit}}^{\tau=0.25}$	$T_{\text{exit}}^{\tau=0.5}$	$T_{\text{exit}}^{\tau=0.75}$	$T_{\text{exit}}^{\tau=0.95}$
0.1	22.0	22.0	23.0	27.0	35.0	34.0	29.0	29.0
0.5	22.0	22.0	23.0	27.0	35.0	34.0	29.0	29.0
1.0	22.0	22.0	23.0	27.0	35.0	34.0	29.0	29.0

Table 11: Entry and exit times based on GAL loss, for each (τ, λ) for France, using thresholds $\varepsilon = 5$ and $\varepsilon' = 1$.

λ	$T_{\text{entry}}^{\tau=0.25}$	$T_{\text{entry}}^{\tau=0.5}$	$T_{\text{entry}}^{\tau=0.75}$	$T_{\text{entry}}^{\tau=0.95}$	$T_{\text{exit}}^{\tau=0.25}$	$T_{\text{exit}}^{\tau=0.5}$	$T_{\text{exit}}^{\tau=0.75}$	$T_{\text{exit}}^{\tau=0.95}$
0.1	81.0	78.0	74.0	87.0	88.0	87.0	86.0	89.0
0.5	81.0	78.0	74.0	87.0	88.0	87.0	86.0	89.0
1.0	81.0	78.0	74.0	87.0	88.0	87.0	86.0	89.0

Table 12: Entry and exit times based on GAL loss, for each (τ, λ) for Germany, using thresholds $\varepsilon = 15$ and $\varepsilon' = 2$.

λ	$T_{\text{entry}}^{\tau=0.25}$	$T_{\text{entry}}^{\tau=0.5}$	$T_{\text{entry}}^{\tau=0.75}$	$T_{\text{entry}}^{\tau=0.95}$	$T_{\text{exit}}^{\tau=0.25}$	$T_{\text{exit}}^{\tau=0.5}$	$T_{\text{exit}}^{\tau=0.75}$	$T_{\text{exit}}^{\tau=0.95}$
0.1	53.0	53.0	53.0	55.0	69.0	71.0	70.0	70.0
0.5	53.0	53.0	53.0	55.0	69.0	71.0	70.0	70.0
1.0	53.0	53.0	53.0	55.0	69.0	71.0	70.0	70.0

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